Lecture 7. Innovation. Industry Life cycle. Disruption

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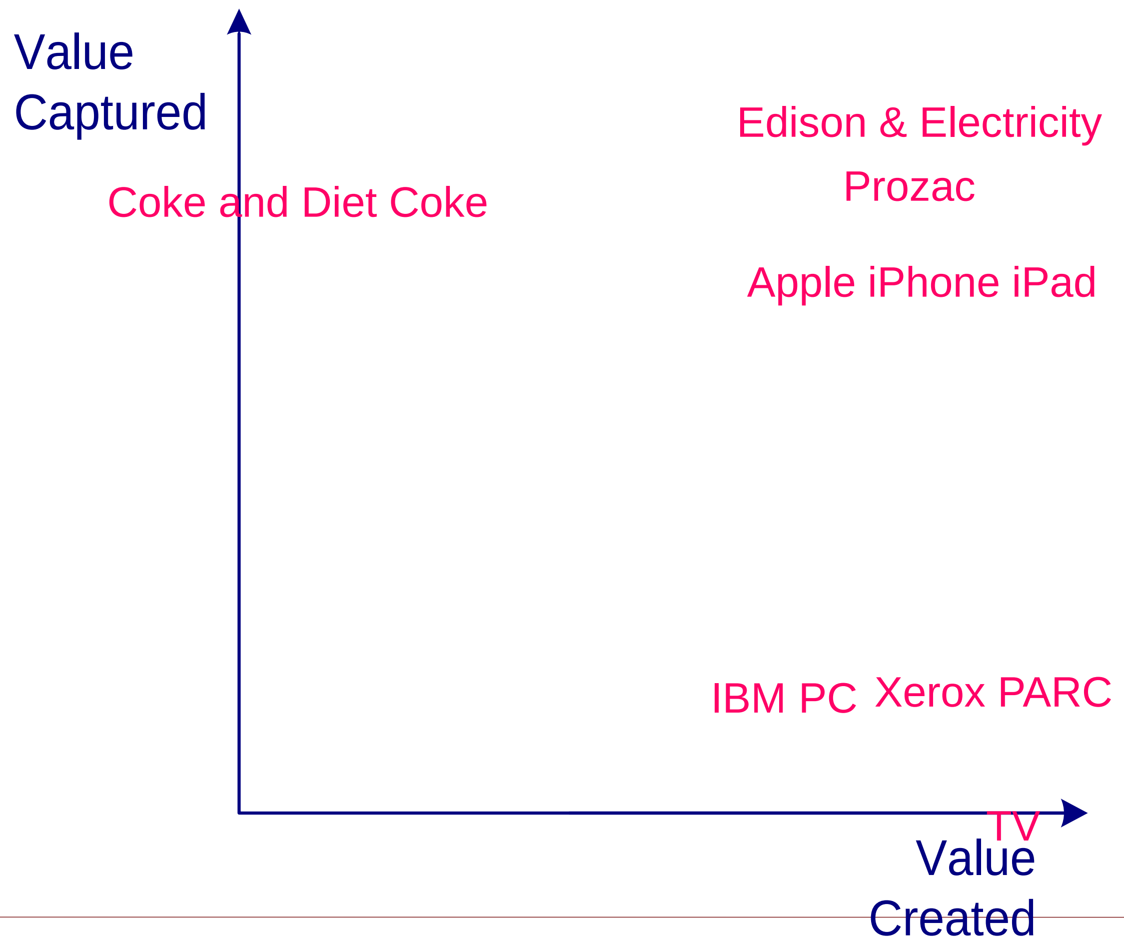
Great Ideas = Pots of Money ?



Outline

I. Does the innovation create value?
   - Forecasting size and composition of demand

II. The challenge for innovator
   1. Short run capture
   2. Long run advantage

III. The challenge for the incumbent:
   - Reorienting strategy to deal with disruption

I. Understanding Innovation Value and Market Dynamics

Challenge of Demand Prediction

If innovation really adds value, it should be straightforward to identify the value that is created.

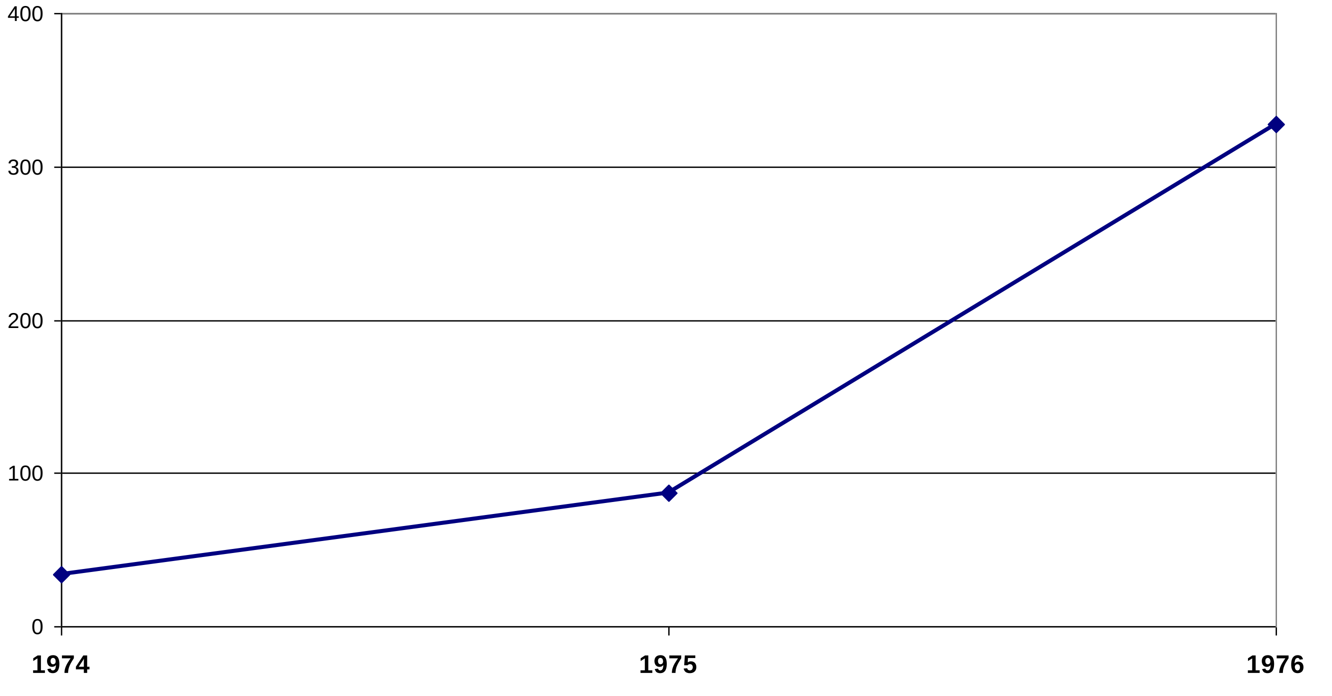
Sadly, it is not

Even when the innovation is immensely valuable

Case Study: CT Scanner

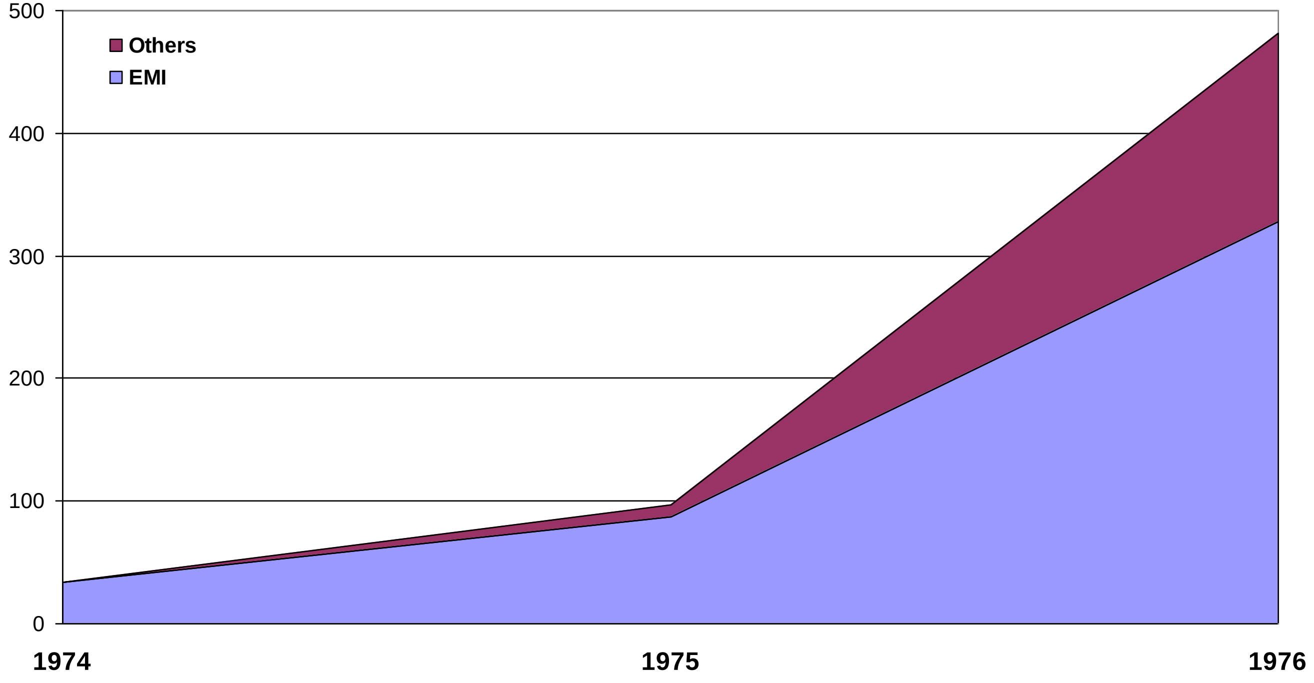
- Illustrates the difficulty of forecasting demand even when tremendous value is created.
- Can be hurt also by competitors' failures to forecast demand (assuming no coordination).

CT Scanner Shipments



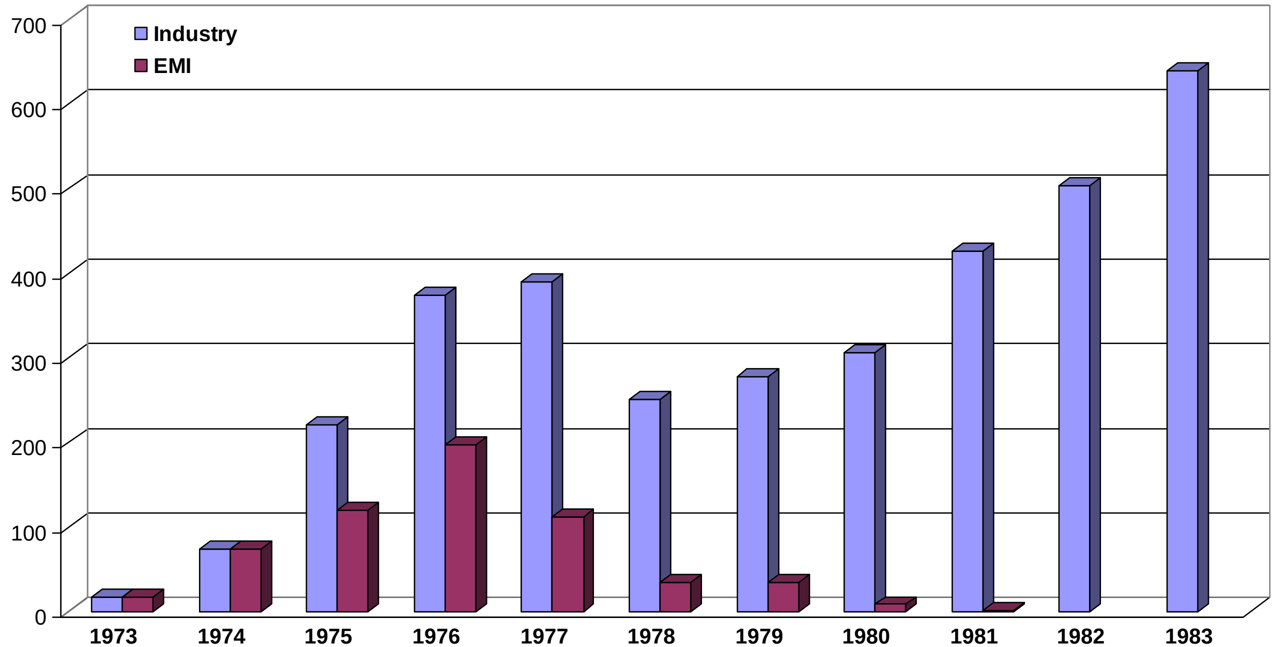
Source: Trajtenberg, 1990, p. 92-101

CT Scanner Shipments (EMI versus others)



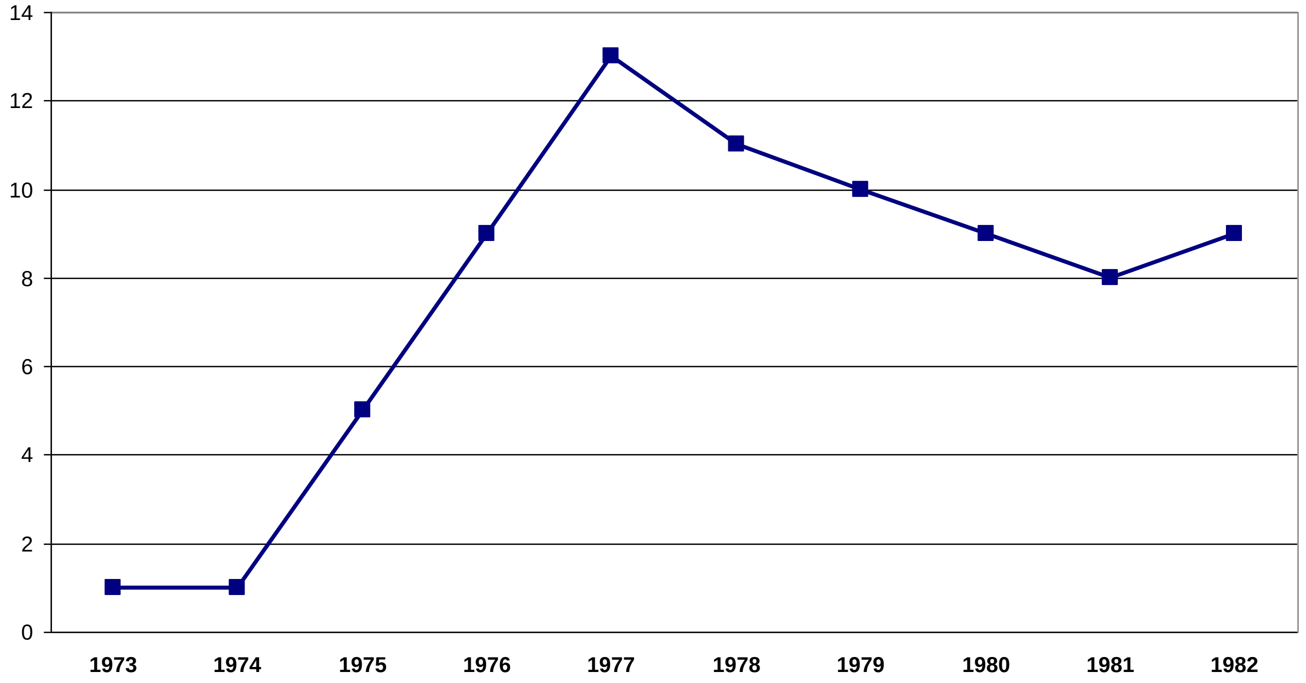
Source: Trajtenberg, 1990, p. 92-101

Units Ordered: U.S. Hospitals 100+ Beds



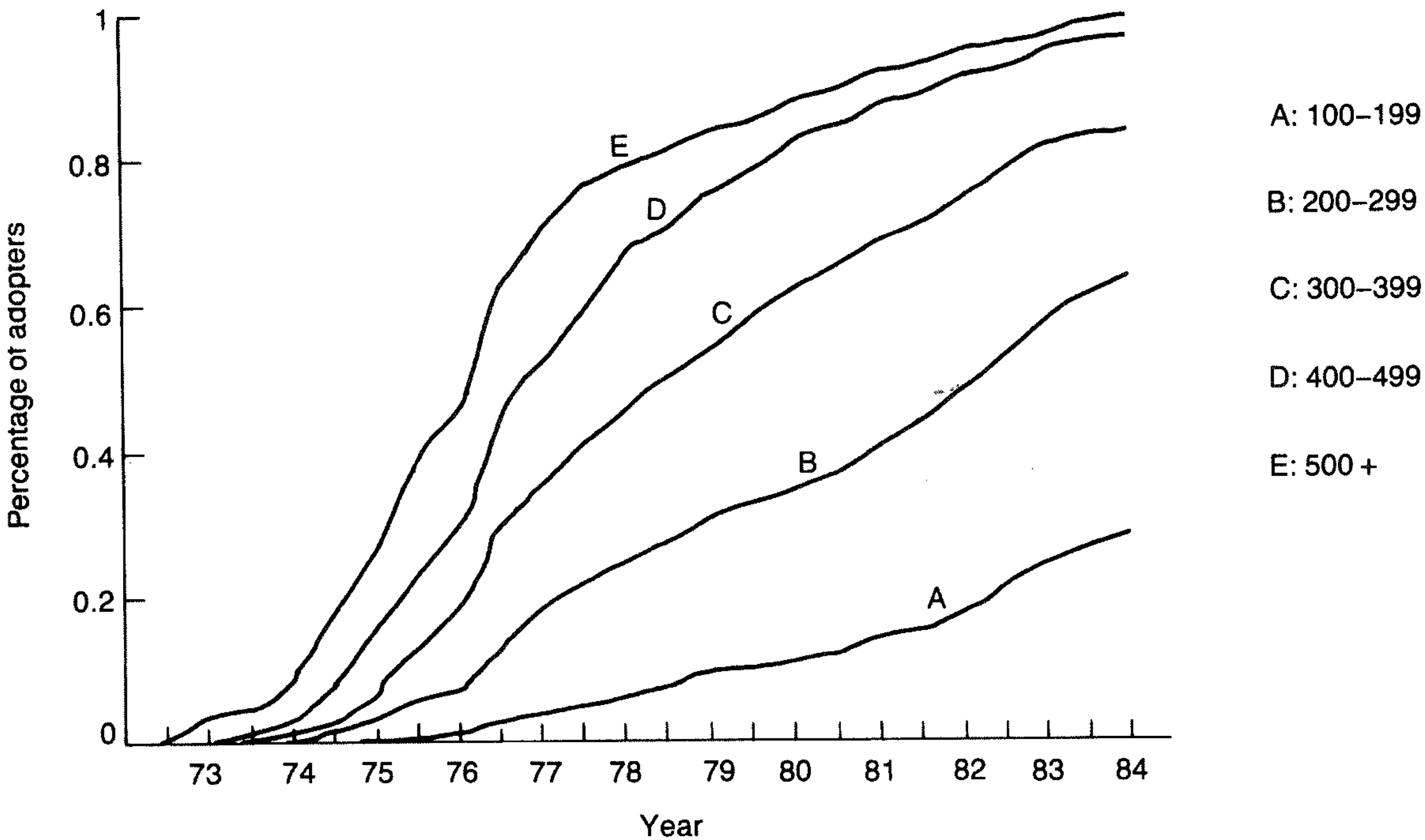
Source: Trajtenberg, 1990, p. 92-101

Number of CT Scanner Suppliers



Source: Trajtenberg, 1990, p. 92-101

Diffusion of CT Scanners by Hospital Size



 Source: Trajtenberg, 1990, p. 59

EMI & the CT Scanner

U.S. demand dropped dramatically. Saturation level of 6/million people nearly reached by 1981

Major price competition with 12 competitors at the peak — rapid shakeout thereafter

Hounsfield receives Nobel Prize in 1979

EMI taken over by Thorn Electric in response to cash crisis

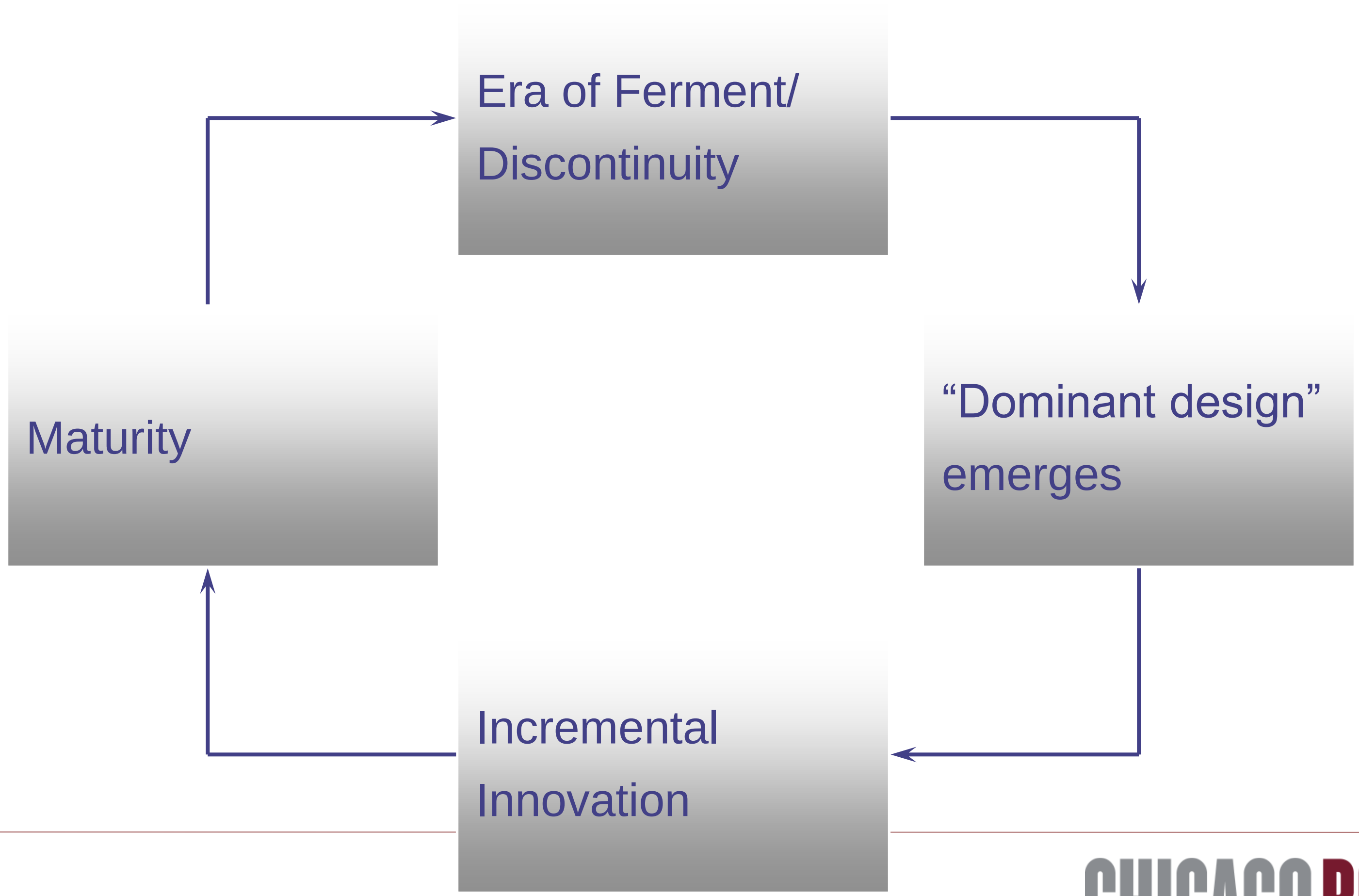
GE buys Medical Division

Key Issue

Forecasting the path of technological change

- Will the new technology take off? How fast? Will it substitute for existing or be complementary with it?

A Key Forecasting Framework: Industry life cycle



Source: Rebecca Henderson

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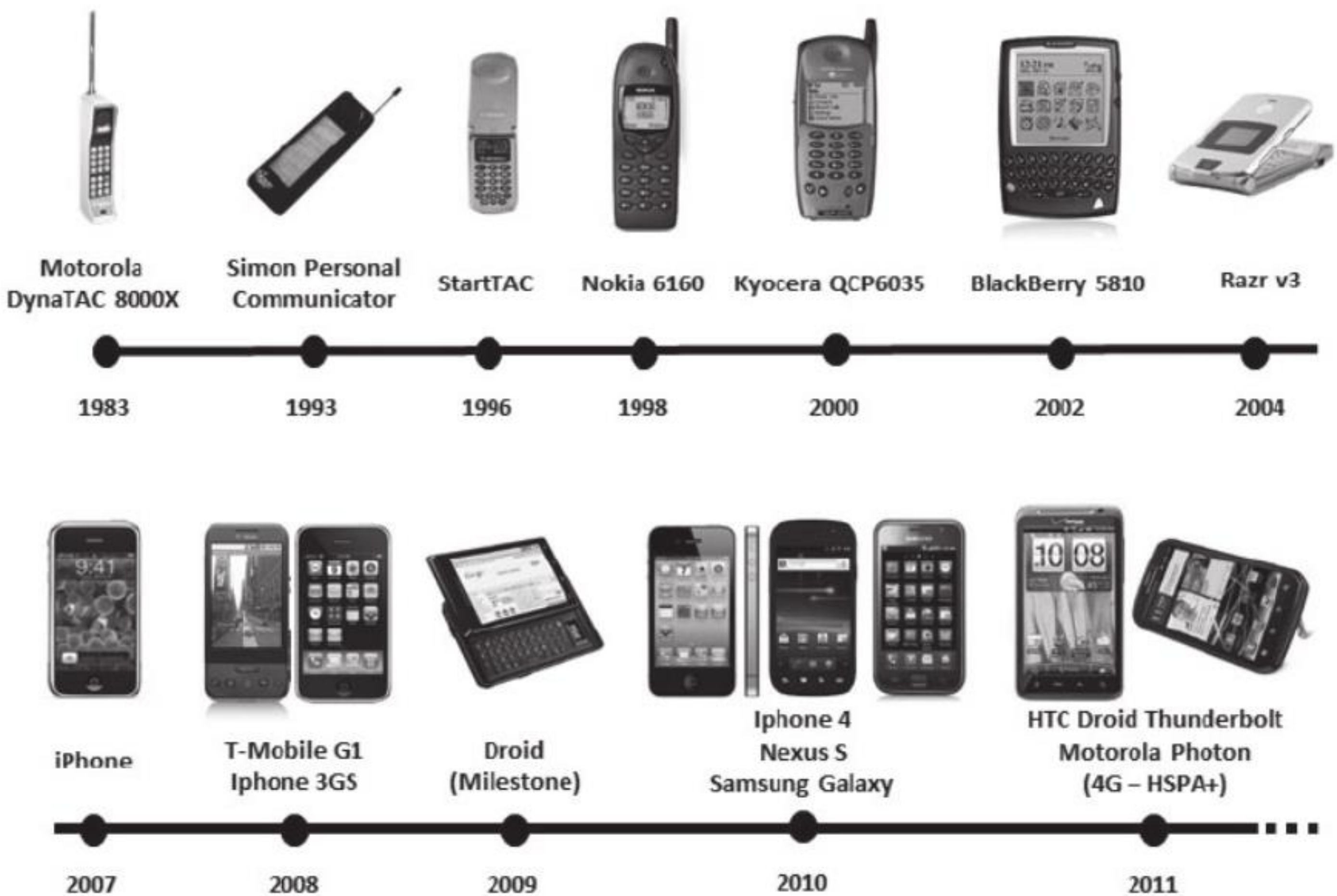
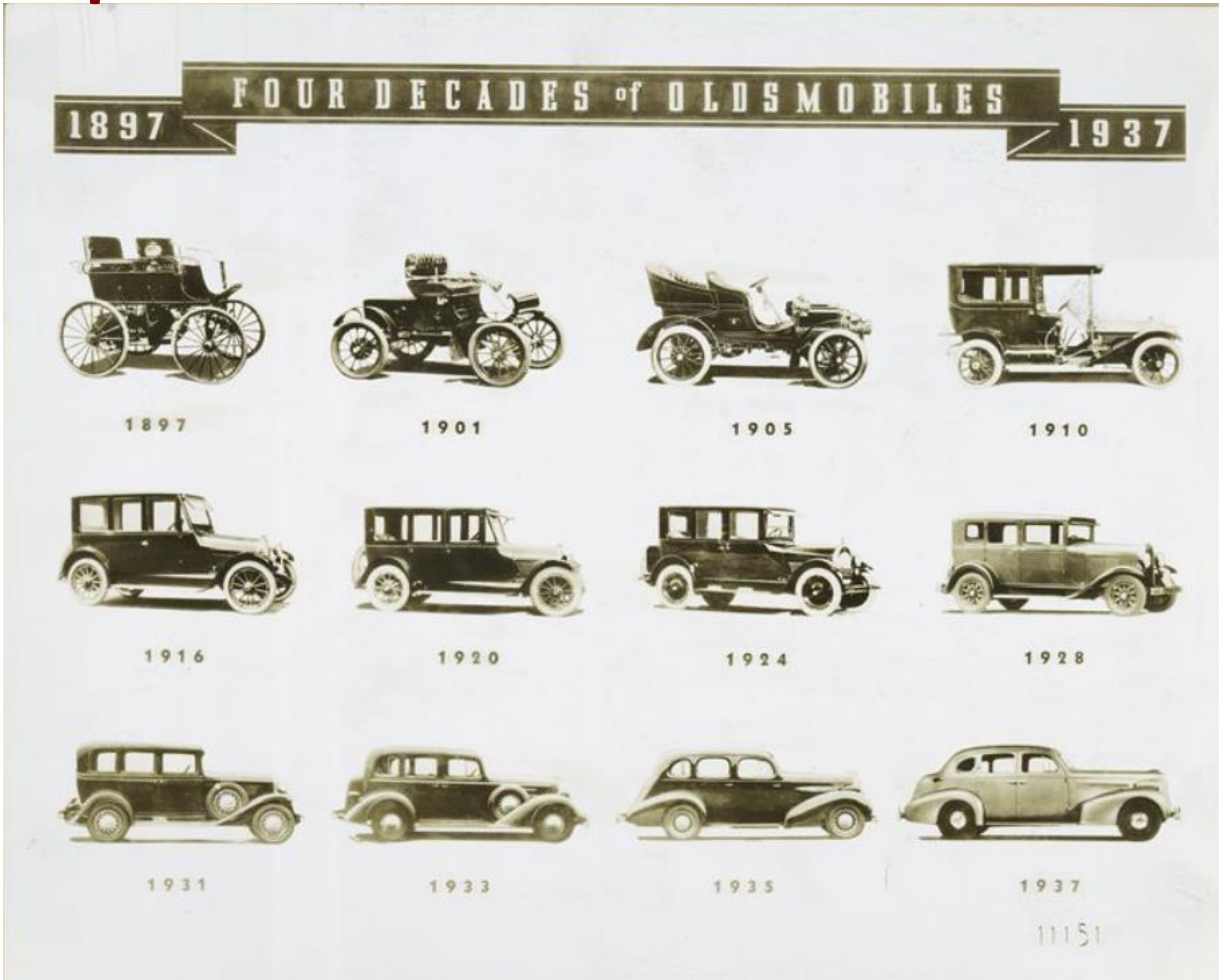
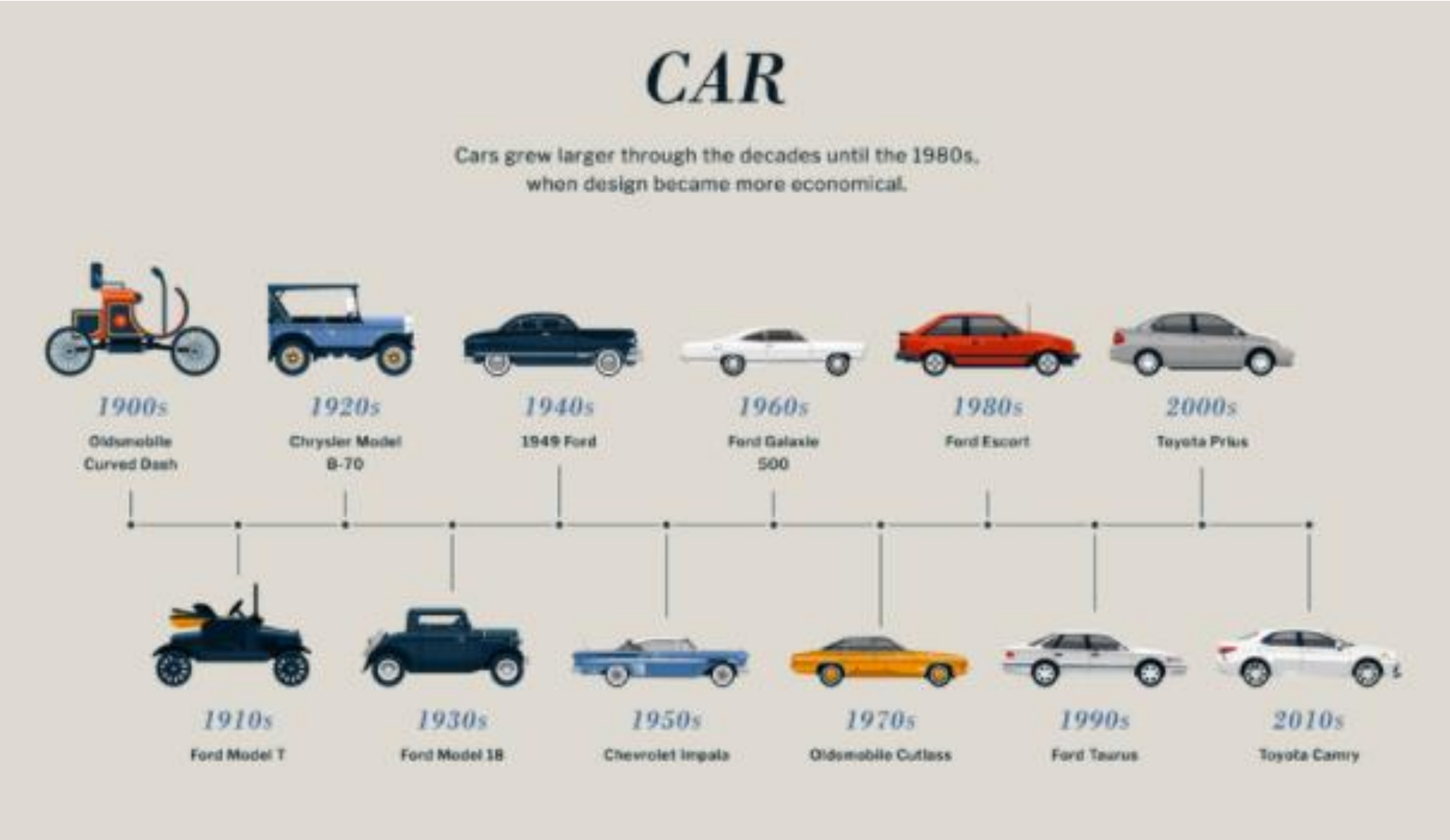
Example 1: Smartphone



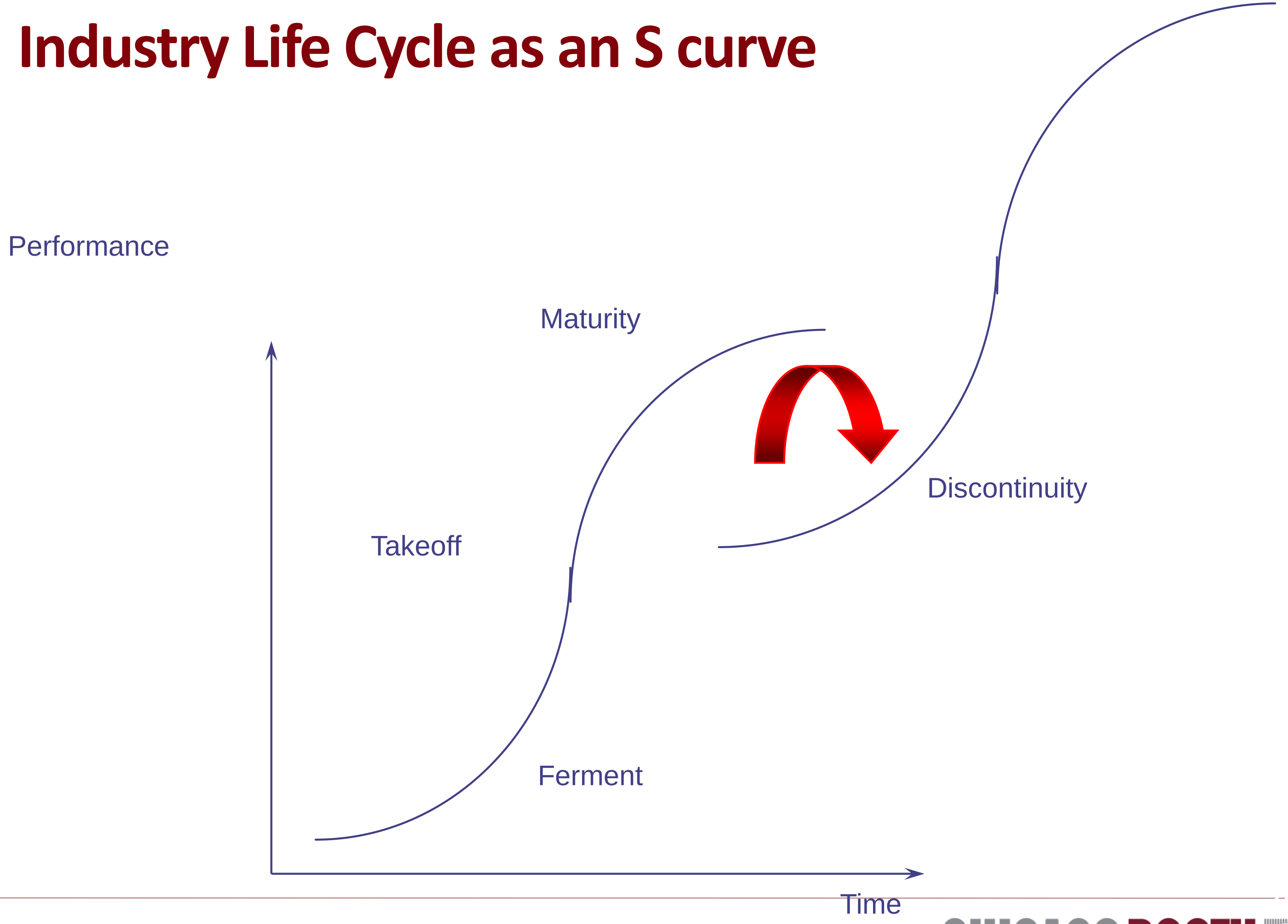
Example 2: Car



Example 2: Car



Industry Life Cycle as an S curve

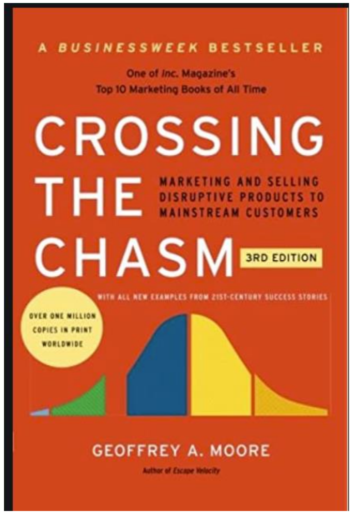


The chasm (Geoffrey Moore) (1991, 2014)

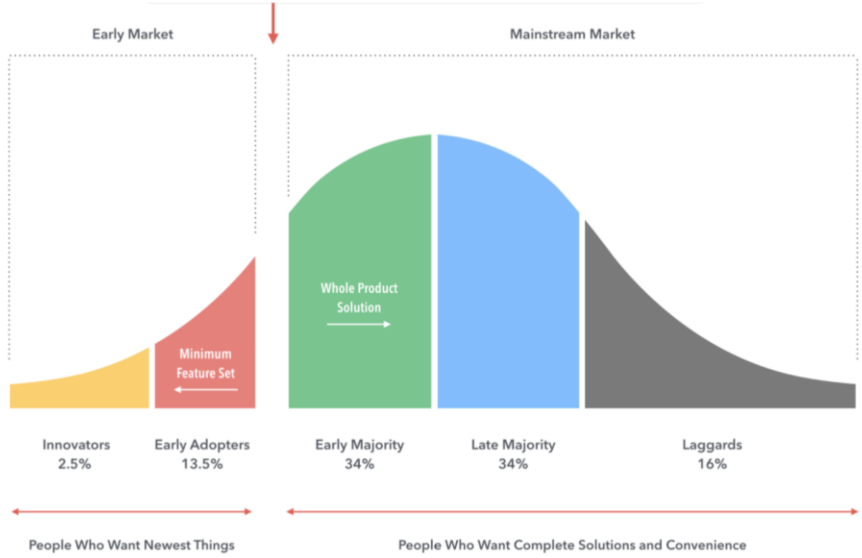
Huge gap between early adopters and "majority"

"Enthusiasts": want something new, want to play with it

"Pragmatists": want something that works



Crossing the chasm



Managerial Implications

Diffusion Paths

- Are very difficult to predict
- The “Chasm” is no legend!

Pioneer innovators face difficulties riding the diffusion curve

- Commercializing in-house is difficult
- Firm faces greatest competition precisely when adoption ‘takes-off’
- Sustaining leadership requires development of systems that entrants find difficult to copy

II. The innovator's problem:
How to capture value from a valuable innovation
in the short run
and long run

Capturing Value From an Innovation

Short run appropriation

Converting innovation into long run advantage

A. Short run capture

Two obstacles

- Difficulty of trading (selling, licensing), and hence, appropriating, IP.

- Need for complementary capabilities or positional resources (rents go to those who have them).

Difficulty of Trading IP vs. Physical Assets

Characteristics	*Know-how/IP*	*Physical Commodities*
Disclosure of Attributes	***Relatively difficult***	***Relatively easy***
Property Rights	***Limited***	***Broad***
Item of Sale	***License***	***Measurable unit***
Variety	***Heterogeneous***	***Homogeneous***
Unit of Consumption	***Often unclear***	***Weight, volume, etc.***
Inherent tradability	***Low***	***High***

Implication: As EMI illustrates, it may be difficult to capture the value from licensing the idea.

Potential Sources of Appropriability

Intellectual property protection

- Patents
  - Finite length
  - The right to prohibit “producing”
- Copyrights
  - The right to prohibit “copying”

Secrecy

- Trade secrets & non-compete clauses
- Complexity and “tacit” knowledge

Speed

Intellectual Property Protection

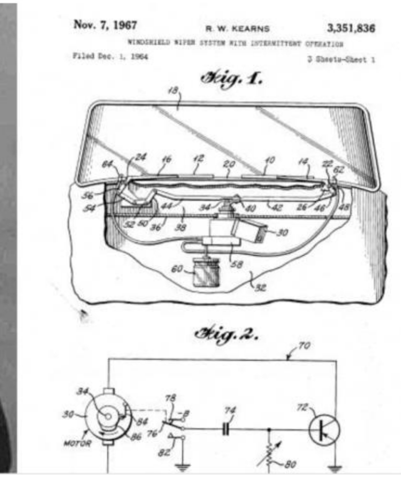
Strengths

- Legal right
- Can be traded
- Buys time to build complementary assets
- Provides temporary monopoly
- Slows competitors down

Weaknesses

- Disclosure requirements
- Costly to enforce
- Can be invented around
- Could be too short
- Not everything can be patented
- False sense of security

Windshield Wipers



1962: Robert Kearns invented a little wiper switch that made the intermittent wiper possible

- Fitted car with it and drove it to Ford
- Ford passed on innovation
- Kearns obtained patents

1969: Ford and others cracked secret

1990: Kearns wins suit against Ford for potential damages payout of $ 325 million (eventually gets $ 8 million).

Secrecy

Strengths

- No disclosure

Weaknesses

- Difficult to maintain
- Non-compete clauses are costly to enforce
- Good technical people do not want their work shrouded in secrecy

Speed

Strengths

- Competitors cannot catch up
- Costly to imitate
- Quick profits

Weaknesses

- Over soon
- Diminishing returns
- Difficult to sustain
- Difficult to pull off/Treadmill
- Dissipates industry profits

The Profits Often Go to the Owners of Complementary Assets

Complementary assets are those necessary to translate an innovation into commercial returns

Positional resources

- Brand name
- Distribution channels
- Customer relationships

Capabilities

- Manufacturing capabilities
- Sales and service expertise

Complementary Assets and Survival in the Typesetter Industry

Waves of innovation

- 1440: manual, Gutenberg
- 1886: ‘hot metal’ linotype machine, Mergenthaler
- 1949: analog phototypesetting
- 1965: digital CRT phototypesetting
- 1976: laser imagesetting

One firm, Mergenthaler Linotype,
survived as industry leader

Mergenthaler's success

1895: recognized need for new font development

1902: library of over 100 fonts

1913: 1000 typefaces

1923: 200 typefaces

Would take 20 years for an entrant to duplicate (with computers, it took Compugraphic a decade and $23 million to generate 1000 fonts)

Key fonts trademarked: "Helvetica"

Complementary assets as a source of market power.

Assess Control of Complementary Assets

Suppose that innovation had been developed by an "external" start-up team

- Would the start-up consider you the ideal partner?
- Are there any capabilities for which it is necessary to approach a partner? A potential competitor??

Firm controls stage-specific assets if a start-up would consider you the ideal partner and there are no "bottleneck" issues

- EMI in Scanners?

B. Long run competitive advantage

Can the innovator convert the idea into a long run advantage?

In Science, yes: glory goes to the first to make the discovery

E.g. Hounsfield

Yet, not so clear in the race to commercialize innovations.

- Second movers may have an advantage
- May prefer to harvest value rather than reinvest.

Innovator Advantages

Market power until entry

Potential long-run advantages

- Positional
  - Reputation, brand, switching costs
  - Economies of Scale
- Capabilities: Learning curve,
- Resources
  - location, distribution channels, other inputs
  - patents

These advantages require investment!!

- Often an important tradeoff between maximizing value of pre-entry market power and long-run advantage.

Innovator's disadvantages

Also some first-mover disadvantages :

- Technological competition
  - Provision of a public good.
  - Non-proprietary technology/ Patent protection?
    - Can new technology be leapfrogged
- Uncertainty on market size and structure
  - Entry/Future market structure
  - Resolving demand uncertainty
- Regulatory approval

III. Disruption and the incumbent's problem: How to Reorient Strategy in the Face of Technological Discontinuity

Disruption

Established firms can be vulnerable to disruption:

1. Demand side disruption
2. Supply side disruption
3. Fear of cannibalization

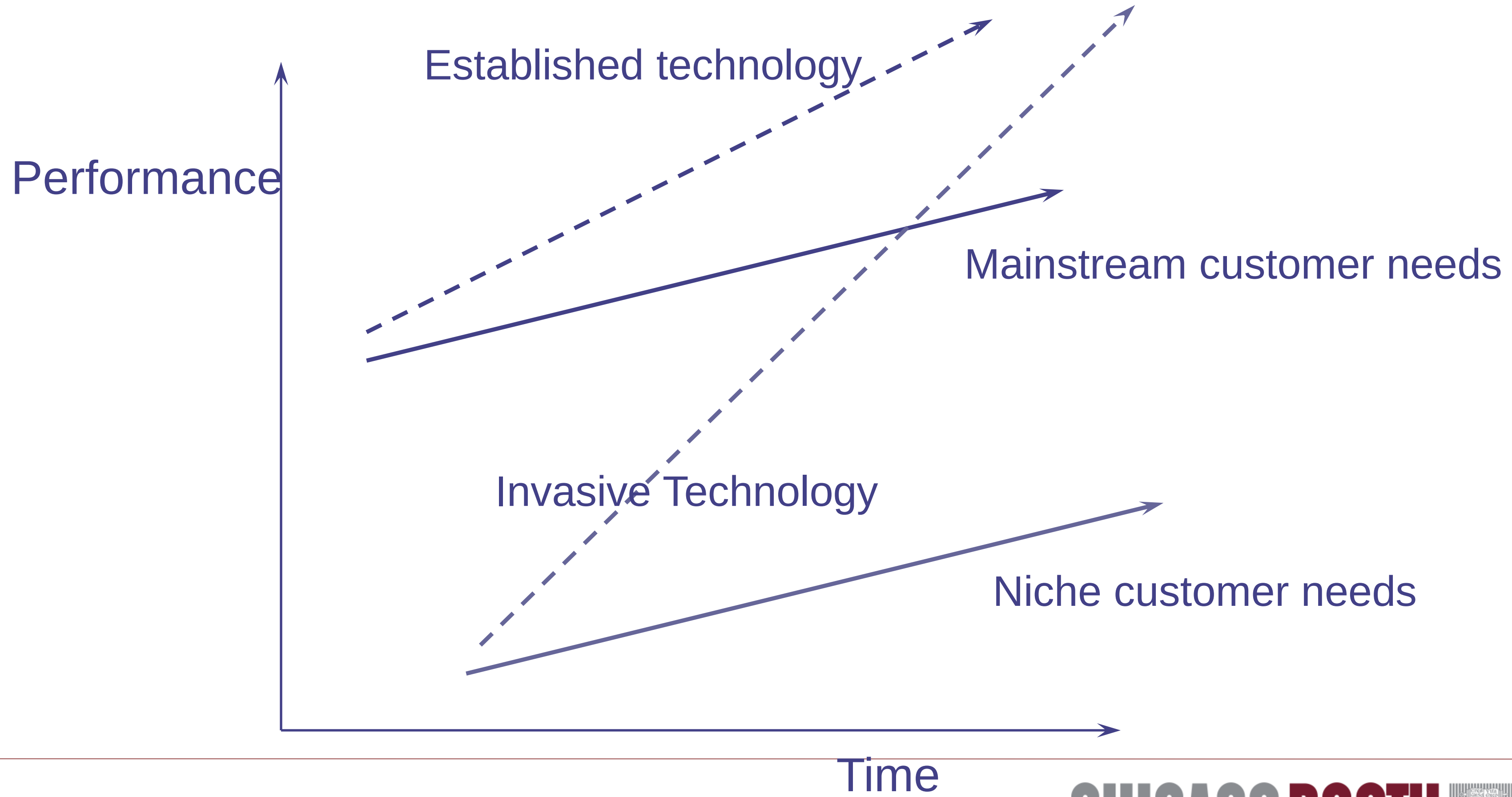
Demand side disruption

Innovation affects customer demand patterns.

Problematic Innovations: lower-cost, accessible products or services that appeal initially to less-demanding customers, eventually capturing a larger market share as their offerings improve.

- Disruptive innovations initially target niche markets or overlooked segments
- E.g. PCs or Digital photography initially offering lower quality but more convenient and accessible options

Christensen: Staying Close to Your Customers May Be a Problem



Managerial Implications

- Explore emerging trends and unmet needs within niche markets or less-demanding customer segments

- Develop and test lower-cost, potentially lower-margin innovations that target new or underserved markets to capture and grow these segments before competitors.

- Understand and segment customers more finely to identify those who might benefit from simpler, more accessible products or services, guiding targeted innovation efforts.

- Acquire new entrants or “disrupt yourself” by setting up new units (more later)

2. Supply side disruption

The way the product is put together changes in way that are incompatible with the organization;

More broadly, the organization/architecture required by the innovation is incompatible with the existing architecture

Problematic innovations: architectural innovations that change the way components of a product are linked together can be difficult for firms to adopt due to established processes (Henderson and Clark)

- leading to potential failure even in the face of technologically inferior competitors.
  - E.g. Transition from mechanical to electronic watches
  - E.g. Blackberry vs iPhone

Henderson and Clark's framework

Henderson and Clark distinguish two types of innovation.

- Component innovation improves parts without altering overall design.

- Architectural innovation rearranges the links among components without changing the parts themselves.

This shift in configuration redefines product functionality and market expectations

The trouble with architectural innovation

- Incumbents struggle to reconfigure routines and asset complements optimized for incremental component improvements.

- New entrants exploit these gaps by designing systems from the ground up.

Managers must reassess internal structures and develop flexible capabilities to adapt to new architectures

A complicated transition to the future

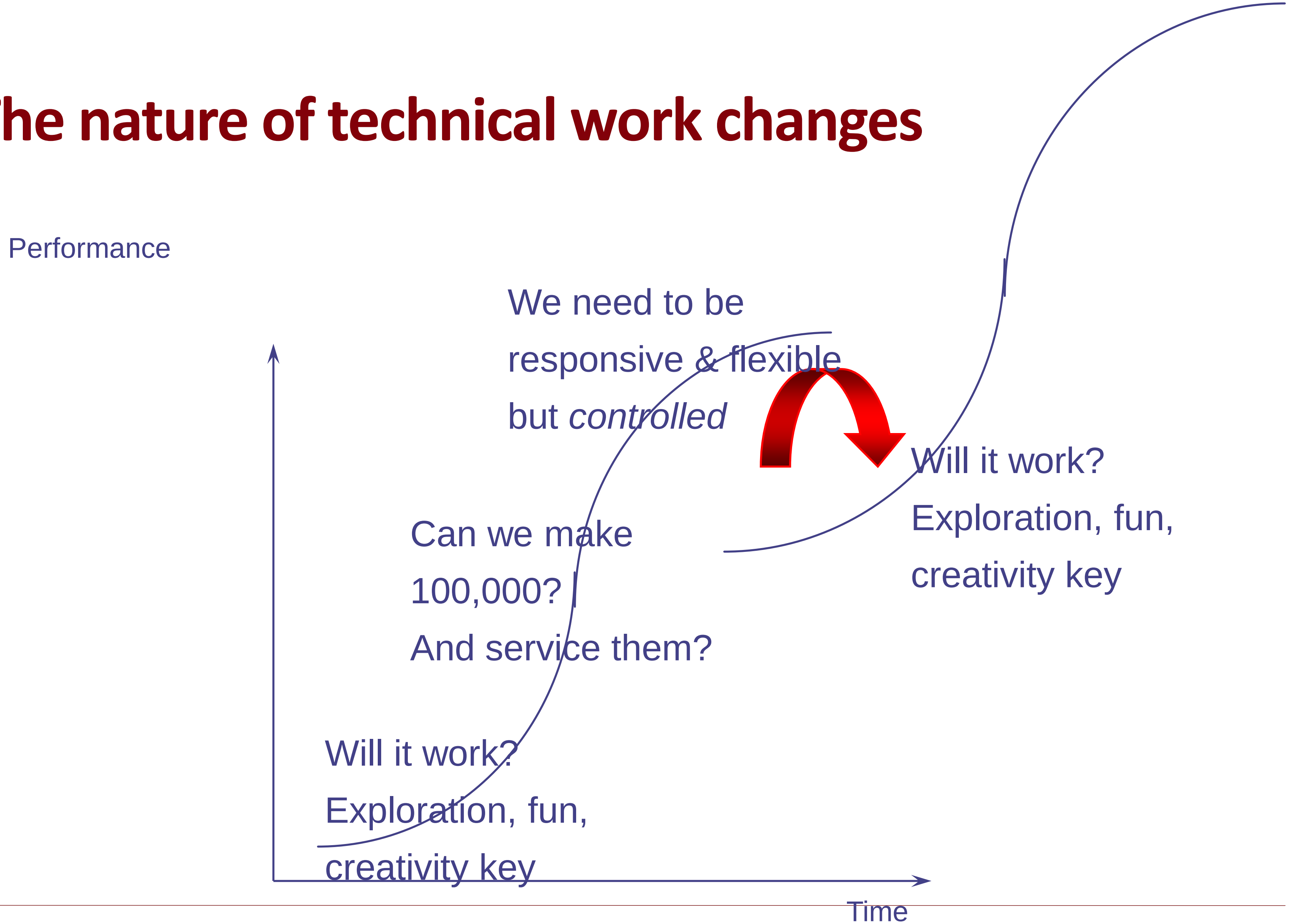
Managing the transition

- Leveraging existing capabilities and positional assets to
  - Create a new market position
  - Develop new organizational capabilities

Problem:

- Capability and positional assets necessary for the new technology may be fundamentally at odds with those that were complementary with the existing

The nature of technical work changes



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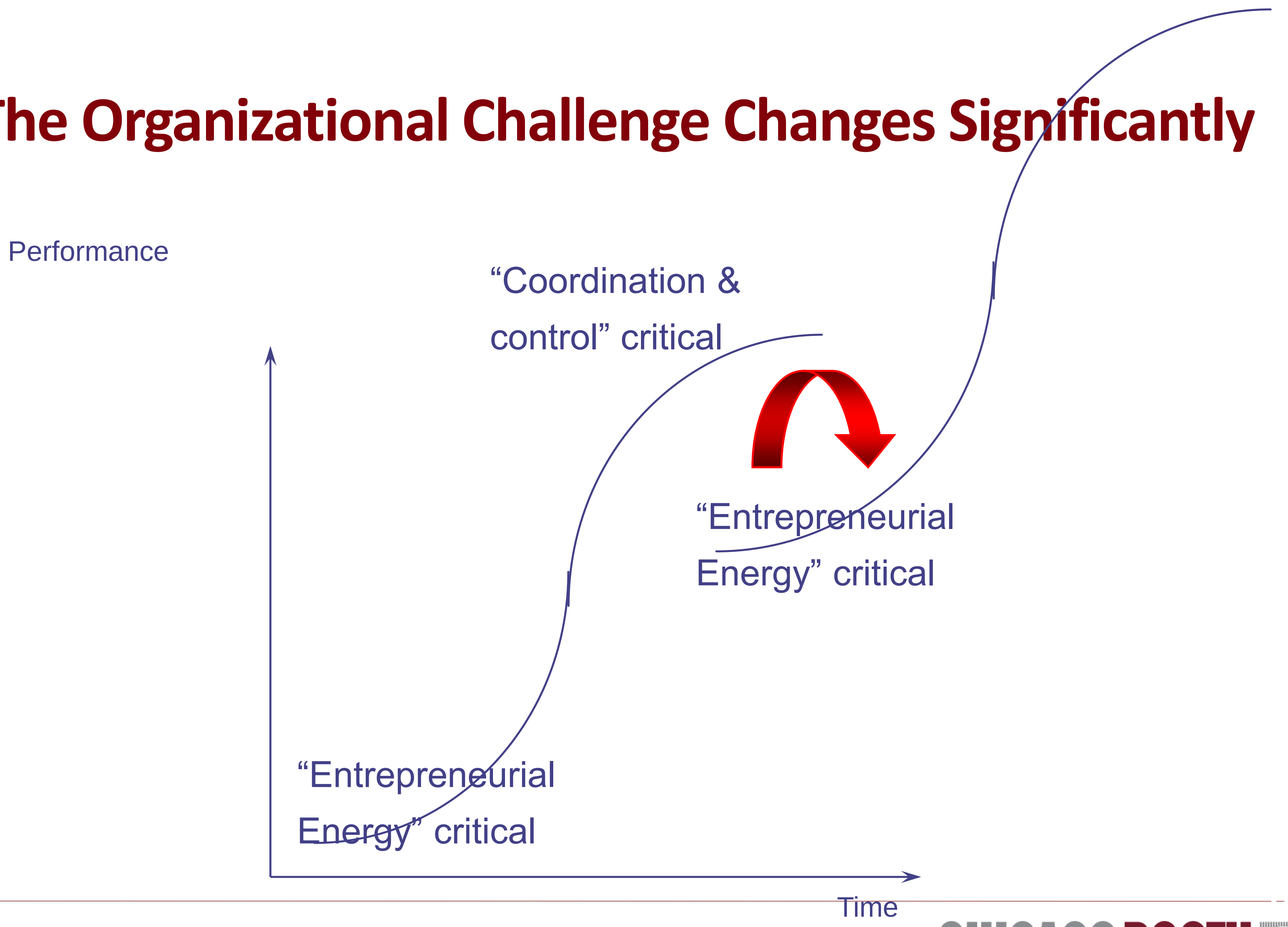
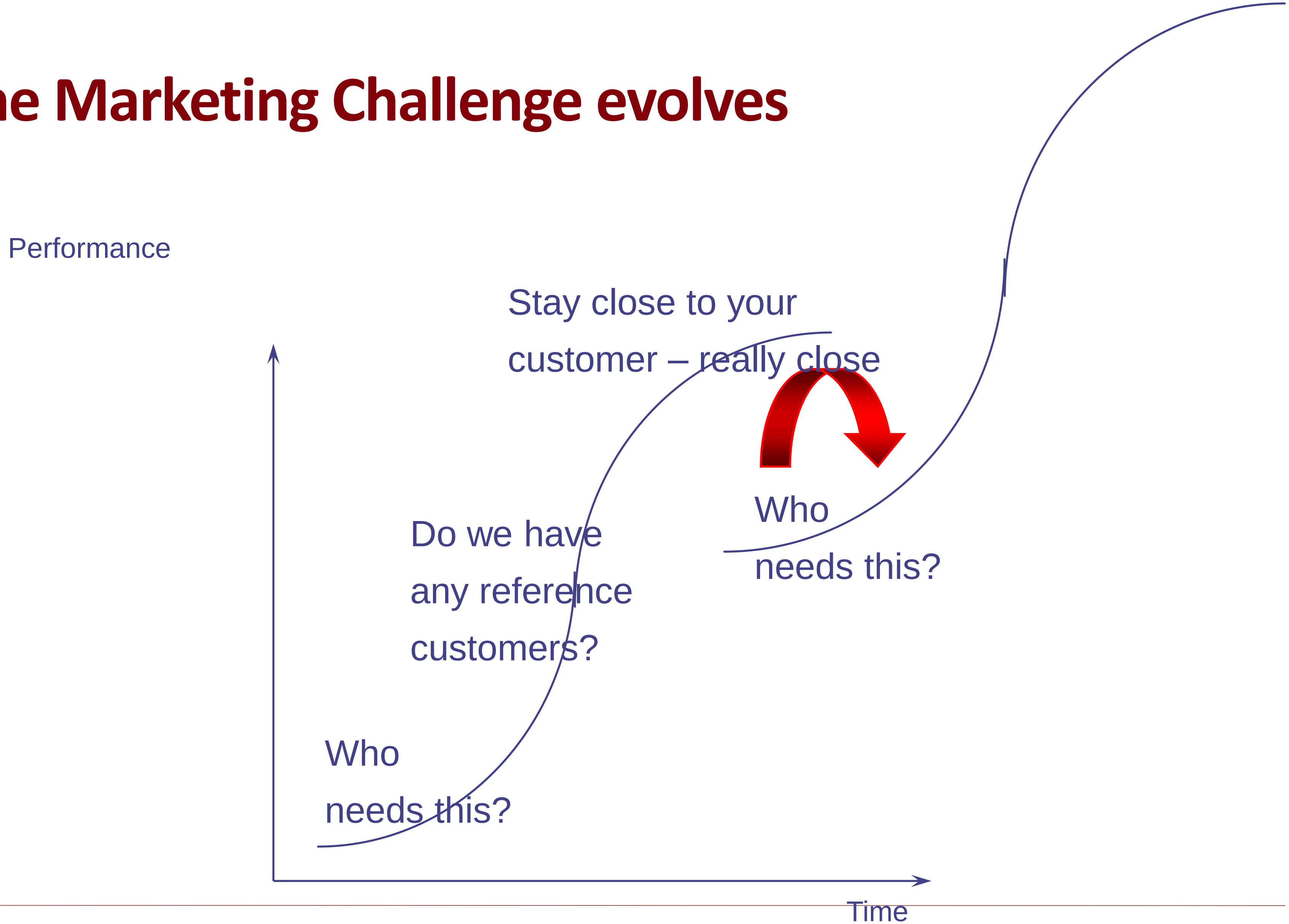
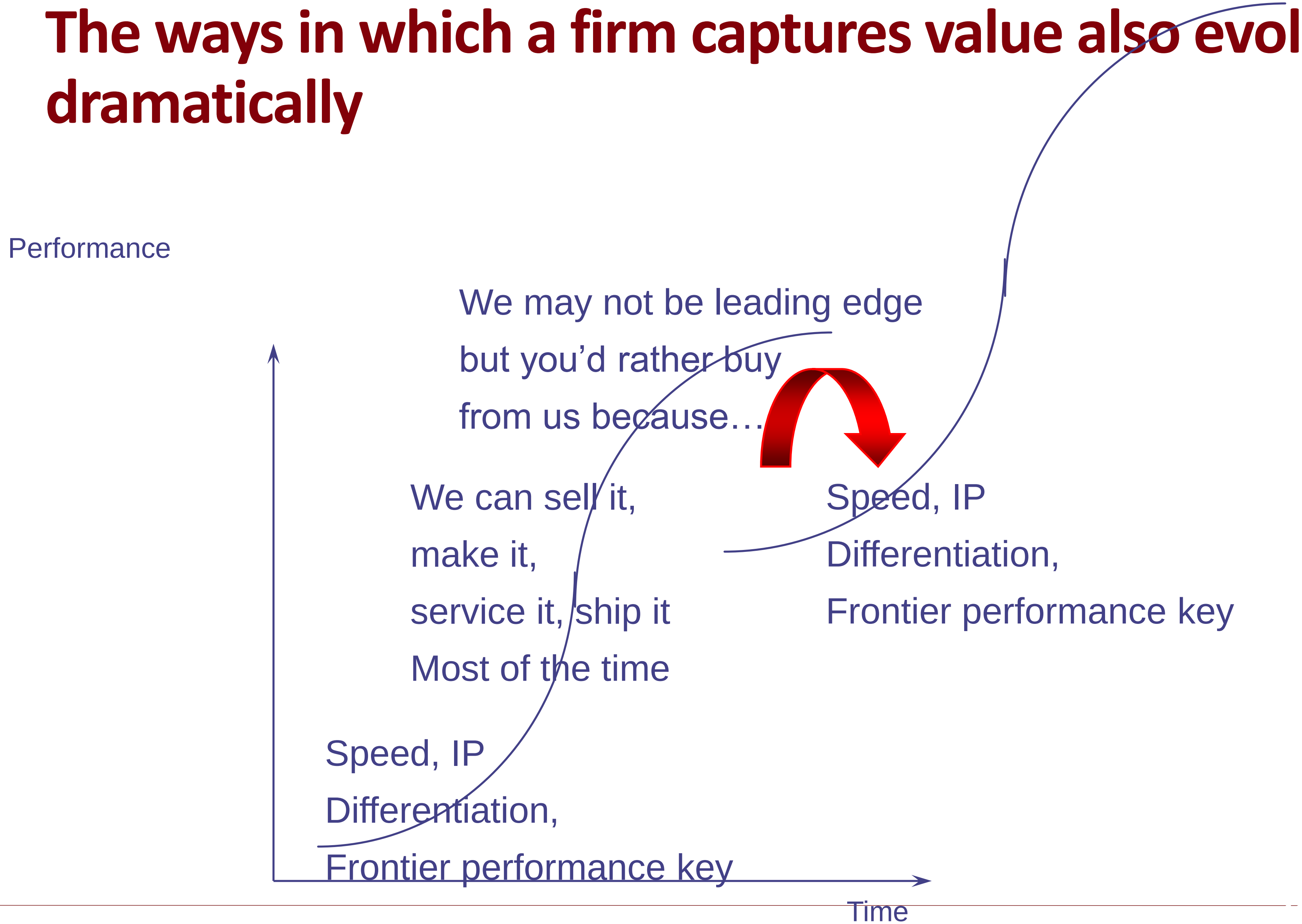
The Organizational Challenge Changes Significantly



The Marketing Challenge evolves



The ways in which a firm captures value also evolve dramatically



Managerial Implications

- Promote agility, cross-functional collaboration, and the ability to respond quickly to technological changes

- Create or participate in ecosystems that foster innovation through partnerships, collaborations, and investments in startups

- Provide incentives that encourage (controlled) risk-taking and experimentation even when it challenges the status quo.
  - But—risk to your brand?

3. Cannibalization fear

Incumbent may be afraid of accelerating cannibalization

- Incumbents fearing revenue loss from new innovations may delay or ignore technological advances risking market disruption

Example: Why did Google lose its advantage on AI?

AI innovations could disrupt its main revenue stream

Innovator does not have that fear

Risk that protecting short term revenue flow leads to long-term strategic disadvantages

How to deal with disruptive technologies?

Internal Startups and Spinoffs

Corporate Venture Capital

Remote location/Separate Business

Internal Start Ups and Spinoffs

Create quasi-independent entities from a parent company to pursue product development or market opportunities that do not align completely with the parent company's core business.

- Internally: Can operate with greater focus and agility than their parent company, pursuing innovative products or services

- Externally: Can test new markets and business models in a relatively safe manner, with the potential for reintegration or further independence based on success

Corporate Venture Capital

Invest in equity stakes in external startups

Also provide management and marketing expertise.

Advantage

- Explore new technologies and business models without the constraints of core operations.

Challenge

- Integration with core strategy

Example: Intel Capital

Objective: develop Intel ecosystem

Remote Location (the PARC model)

Establishing operations in remote locations

Advantage:

- Access to New Ideas:
- Cultural and Operational Flexibility

Challenge: integrating the innovations

Kodak's failure

- Early days (1983-1993)
  - Massive R&D investments in digital imaging, despite the absence of a market ($ 5BL)
    - Many technical achievements
    - Camera available in early nineties
  - Product market strategy driven by desire to replicate traditional "razor-blade" model
    - Despite technological achievements, attention focused on Photo-CD
- Fisher's era (1993-1997)
  - Radical strategic change: we are an imaging company
    - Making money on blades as well as razors
  - Attempted radical intra-organizational change
    - Refocusing on core; Rationalization of operations; Reorganization; Managerial change
  - Middle managers resist new strategic orientation

Kodak

- While the company accumulates losses in digital imaging, Fuji attacks core business…
- Core business:
  - Dramatic loss of market share

Worldwide:	1983	1997	2001
Kodak	80%	46%	36%
Fuji	8-10%	29%	37%

US:	1983	2001
Kodak	90%	60%
Fuji	7-8%	38%

  - Fuji's increased penetration, relationships with retailers
    - A massive failure

Kodak, lessons

- The disruption may actually have huge costs on the current business
  - Kodak's massive R&D investments in digital imaging despite the absence of a market ($ 5BL) and the likelihood that the traditional business had considerable staying power.
  - Together with ill-advised diversification, may have dissipated resources and attention from responding to Fuji.

Kodak, lessons

- Kodak illustrates the difficulty of trying to change from within…
  - Difficulty of overcoming entrenched culture and strategy reflected in poor execution of the Photo-CD, fixation on “razor-blade” model.
  - In general, difficulty of instituting change from within leads to the oft-heard advice to separate new ventures in “greenfield” divisions (e.g., Saturn).

Kodak, lessons

- … but change from “without” may be difficult as well.
  - If digital had been done “far from Rochester,” what would Kodak have been leveraging? And if the goal is to change the core of the organization, doing it with a greenfield operation may hinder this (e.g., Saturn again).
  - While Fisher scored some major successes (rollback of misguided diversification, China), he encountered difficulty getting buy-in from middle-managers.

Wrap up

Take aways for innovators

Take aways for incumbents

Internal and external aspects

Lessons for innovators

Strategic Protection

- Develop IP strategy
- Acquire complementary assets
- Market timing (see EMI!)

Market Development

- Customer segment targeting
- Channel development
- Ecosystem building

Lessons for incumbents

Organizational Adaptation

- Structural flexibility
- Resource reallocation
- Culture transformation

Strategic Balance

- Core business maintenance
- Innovation investment
- Risk management
  - Technology risk
  - Market risk
  - Organizational risk

Key Ideas and Frameworks

External:

- Technology trajectory (industry life cycle)
  - Industry life cycle
  - S curve
- Demand forecasting
  - Chasm
- Competition
  - Phases – monopoly, entry, shake-out, consolidation

Internal/Capability Evaluation

- Internal resources
- Architectural Innovation
- Complementary assets
- Organizational flexibility/adaptation

In conclusion

1. Enormous difficulty of forecasting demand

   Tools

2. Innovator: Capturing short run and long run value from innovation (IP, Complementary assets)

3. The challenge for the incumbent: the risk of disruption
   - Demand based
   - Supply/organization based

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